Visual Test of typstable

Contents

. 1
`tt()` Base 5
Default (no rownames) 5
rownames=TRUE 5
cols with tidy-select subset 5
col_names custom names 6
align single value 6
align per-column vector 6
col_widths explicit 7
escape=TRUE with special characters 7
escape=FALSE with raw Typst markup 7
Quarto cross-reference 8
`tt_widths()` 8
Positional proportional widths 8
Named column widths 9
Equal explicit widths 9
`tt_style()` 10
stroke=TRUE (full grid) 10
stroke with color 10
striped=TRUE 10
fill (global) 11
inset (global padding) 11
row_gutter with stroke 12
column_gutter with stroke 12
position="center" with full_width=FALSE 12
full_width=TRUE 13
`tt_column()` Static Styling 13
bold=TRUE 13
italic=TRUE 14
color (static string) 14
fill (static) 14
font_size 15
rotate (numeric) 15
rotate (string) 15
inset 16
width override 16

border_left=TRUE . . . 17
border_right with stroke spec . . . 17
`tt_column()` Tidy-Select . . . 18
Multiple columns with c() . . . 18
everything() selector . . . 18
Numeric index . . . 18
`tt_column()` Data-Driven . . . 19
fill via bare column reference . . . 19
color via pattern . . . 19
bold via bare column reference . . . 20
italic via pattern . . . 20
font_size via pattern . . . 21
rotate via bare column reference . . . 21
.missing="ignore" with nonexistent pattern column . . . 21
inset via pattern . . . 22
`tt_row()` . . . 22
Header row styling . . . 22
Single data row fill and color . . . 23
Multiple rows with fill . . . 23
italic and font_size on a row . . . 23
align override on a row . . . 24
rotate on a data row . . . 24
inset on a row . . . 25
hline_above and hline_below . . . 25
`tt_cell()` . . . 25
fill and bold on a cell . . . 25
italic, color, and font_size on a cell . . . 26
align override on a cell . . . 26
rotate on a cell . . . 27
inset on a cell . . . 27
colspan=2 with content override . . . 27
rowspan=3 with content override . . . 28
content override with styling . . . 28
Header cell styling . . . 29
`tt_hline()` and `tt_vline()` . . . 29
Custom hline at position 3 . . . 29
Hline with styled stroke . . . 29
Partial hline with start/end . . . 30
Vline after column 1 . . . 30
Vline with styled stroke . . . 30
Partial vline with start/end . . . 31

`tt_pack_rows()` . . . 31
Single group with default indent and bold . . . 31
Multiple chained groups . . . 32
index parameter . . . 32
indent=FALSE . . . 33
bold=FALSE . . . 34
Styled group labels . . . 34
Group label with color and alignment . . . 35
Basic `tt_header_above()` . . . 36
Single spanning header . . . 36
Multiple groups with default line_sep . . . 36
Groups with empty spacer . . . 37
line_sep=NULL (no separator) . . . 37
line_sep="20pt" (wider separator) . . . 38
Custom styling (bold=FALSE, color, fill, align) . . . 38
line=FALSE . . . 39
italic + font_size . . . 39
rotate . . . 40
inset . . . 40
Stacked Headers . . . 41
Two stacked header rows (no separator) . . . 41
Three stacked headers with different styling (no separator) . . . 41
Two stacked headers with separator on innermost . . . 42
Three stacked headers with separators . . . 42
Targeting Headers with `tt_row()` (Negative Indices) . . . 43
Italic via tt_row(-1) . . . 43
Fill via tt_row(-1) . . . 44
Override bold=FALSE via tt_row(-1) . . . 44
Color + font_size via tt_row(-1) . . . 45
Style different stacked rows independently . . . 45
Three stacked rows with individual styles . . . 46
Targeting Headers with `tt_cell()` (Negative Indices) . . . 46
Fill on one group . . . 46
Color on one group . . . 47
Content override on a group . . . 47
Column normalization: any column in group hits the same cell . . . 48
Style multiple groups independently . . . 48
tt_cell on outer stacked header . . . 49
Content + styling override on stacked header . . . 49
Combined: tt_row + tt_cell on Header Above . . . 50
Row-level style + cell-level override . . . 50

header_spec defaults + tt_row + tt_cell precedence . . . 50
Full example: stacked headers with mixed targeting . . . 51
Precedence / Last-Write-Wins . . . 52
Row after column (row wins) . . . 52
Column after row (column wins) . . . 52
Cell overrides both row and column . . . 53
Non-conflicting attributes merge . . . 53
Data-driven column after row . . . 54
NA Handling . . . 54
NA values in data cells . . . 54
NA in data-driven style columns . . . 55
Complex Compositions . . . 55
Combined alignment with rotation . . . 55
Full-featured table . . . 55
pack_rows and header_above combined . . . 56
Multiple tt_column calls on same column (last wins) . . . 57
colspan and rowspan in same table . . . 57
Color formats . . . 58
Custom pattern . . . 58
`booktabs` Argument . . . 59
Default booktabs=TRUE . . . 59
booktabs=FALSE (grid style) . . . 59
booktabs=FALSE with header styling . . . 60
booktabs=FALSE with header_above . . . 60
`stroke = FALSE` (Line Suppression) . . . 60
Suppress top booktabs rule . . . 60
Suppress mid booktabs rule . . . 61
Suppress bottom booktabs rule . . . 61
Suppress all three booktabs rules . . . 61
Suppress mid rule and add custom replacement . . . 62
stroke=FALSE on individual cells . . . 62
stroke=FALSE on a row in a grid table . . . 63
hline_above=FALSE on a grid table row . . . 63
tt_style(stroke=TRUE) overrides booktabs . . . 63
Override a booktabs rule with custom stroke . . . 64
booktabs=FALSE with vline(stroke=FALSE) . . . 64
styling of tt_pack_rows . . . 65

`tt()` Base

Default (no rownames)

Default table excludes row names.

```
tt(mt)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

rownames=TRUE

Include row names as the first column.

```
tt(mt, rownames = TRUE)
```

	mpg	cyl	disp
Mazda RX4	21	6	160
Mazda RX4 Wag	21	6	160
Datsun 710	22.8	4	108
Hornet 4 Drive	21.4	6	258
Hornet Sportabout	18.7	8	360

cols with tidy-select subset

Select specific columns with tidy-select syntax.

```
tt(mtcars[1:5, ], cols = c(mpg, cyl, hp))
```

mpg	cyl	hp
21	6	110
21	6	110
22.8	4	93

mpg	cyl	hp
21.4	6	110
18.7	8	175

col_names custom names

Override display names for columns.

```
tt(mt, col_names = c("", "Miles/Gallon", "Cylinders", "Displacement"), rownames
= TRUE)
```

	Miles/Gallon	Cylinders	Displacement
Mazda RX4	21	6	160
Mazda RX4 Wag	21	6	160
Datsun 710	22.8	4	108
Hornet 4 Drive	21.4	6	258
Hornet Sportabout	18.7	8	360

align single value

Apply a single alignment to all columns.

```
tt(mt, align = "center")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

align per-column vector

Specify alignment for each column individually.

```
tt(mt, align = c("left", "right", "center"))
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

col_widths explicit

Provide explicit column widths via the `col_widths` parameter.

```
tt(mt, col_widths = c("100pt", "2fr", "1fr"))
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

escape=TRUE with special characters

Typst special characters are escaped by default.

```
special <- data.frame(
  col1 = c("*bold*", "_underline_", "#heading"),
  col2 = c("@ref", "<angle>", "a > b"),
  stringsAsFactors = FALSE
)
tt(special, escape = TRUE)
```

col1	col2
bold	@ref
underline	<angle>
#heading	a > b

escape=FALSE with raw Typst markup

Allow raw Typst markup to pass through unescaped.

```r
raw_typst <- data.frame(
  col1 = c("#strong[Bold]", "#emph[Italic]", "plain"),
  col2 = c("a", "b", "c"),
  stringsAsFactors = FALSE
)
tt(raw_typst, escape = FALSE)
```

col1	col2
Bold	a
Italic	b
plain	c

Quarto cross-reference

Test Quarto `tbl-cap` and `label` options.

```r
tt(mt) |>
  tt_style(stroke = TRUE)
```

Table 1: This is a caption

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

See Table 1.

`tt_widths()`

Positional proportional widths

Set widths as proportional values applied in column order.

```r
tt(mt) |>
  tt_widths(1, 3, 1)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Named column widths

Target specific columns by name.

```
tt(mt) |>
  tt_widths(mpg = 2, cyl = 1, disp = 3)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Equal explicit widths

All columns get equal proportions.

```
tt(mt) |>
  tt_widths(1, 1, 1)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_style()`

stroke=TRUE (full grid)

Add default borders to all cells.

```
tt(mt) |>
  tt_style(stroke = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

stroke with color

Add blue 2pt borders.

```
tt(mt) |>
  tt_style(stroke = "2pt + blue")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

striped=TRUE

Alternate row fill colors.

```
tt(mt) |>
  tt_style(striped = TRUE)
```

mpg	cyl	disp
21	6	160

mpg	cyl	disp
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

fill (global)

Set a fill color for the entire table.

```
tt(mt) |>
  tt_style(fill = "#f5f5dc")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

inset (global padding)

Set cell padding globally.

```
tt(mt) |>
  tt_style(inset = "8pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

row_gutter with stroke

Add vertical spacing between rows.

```
tt(mt) |>
  tt_style(stroke = TRUE, row_gutter = "4pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

column_gutter with stroke

Add horizontal spacing between columns.

```
tt(mt) |>
  tt_style(stroke = TRUE, column_gutter = "4pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

position="center" with full_width=FALSE

Center a non-full-width table on the page.

```
tt(mt) |>
  tt_style(position = "center", full_width = FALSE)
```

mpg	cyl	disp
21	6	160

mpg	cyl	disp
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

full_width=TRUE

Force the table to span the full page width.

```
tt(mt) |>
  tt_style(full_width = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_column()` Static Styling

bold=TRUE

Make the mpg column bold.

```
tt(mt) |>
  tt_column(mpg, bold = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

italic=TRUE

Make the cyl column italic.

```
tt(mt) |>
  tt_column(cyl, italic = TRUE)
```

mpg	*cyl*	disp
21	*6*	160
21	*6*	160
22.8	*4*	108
21.4	*6*	258
18.7	*8*	360

color (static string)

Apply blue text color to a column.

```
tt(mt) |>
  tt_column(mpg, color = "blue")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

fill (static)

Apply yellow fill to a column.

```
tt(mt) |>
  tt_column(disp, fill = "yellow")
```

mpg	cyl	disp
21	6	160
21	6	160

mpg	cyl	disp
22.8	4	108
21.4	6	258
18.7	8	360

font_size

Set a smaller font size on a column.

```
tt(mt) |>
  tt_column(mpg, font_size = "8pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

rotate (numeric)

Rotate column content by 90 degrees using a numeric value.

```
tt(mt) |>
  tt_column(cyl, rotate = 90)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

rotate (string)

Rotate column content by 45 degrees using a string value.

```r
tt(mt) |>
  tt_column(cyl, rotate = "45deg")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

inset

Apply custom cell padding to a column.

```r
tt(mt) |>
  tt_column(mpg, inset = "12pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

width override

Override the width of a single column.

```r
tt(mt) |>
  tt_column(mpg, width = "100pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

border_left=TRUE

Add a left border to a column.

```
tt(mt) |>
  tt_column(cyl, border_left = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

border_right with stroke spec

Add a styled right border to a column.

```
tt(mt) |>
  tt_column(cyl, border_right = "2pt + red")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_column()` Tidy-Select

Multiple columns with c()

Style multiple columns at once.

```
tt(mt) |>
  tt_column(c(mpg, disp), bold = TRUE, color = "navy")
```

mpg	cyl	**disp**
21	6	**160**
21	6	**160**
22.8	4	**108**
21.4	6	**258**
18.7	8	**360**

everything() selector

Apply styling to all columns.

```
tt(mt) |>
  tt_column(everything(), fill = "#f0f0f0")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Numeric index

Target a column by its numeric position.

```
tt(mt) |>
  tt_column(1, italic = TRUE, color = "red")
```

mpg	cyl	disp
21	6	160

mpg	cyl	disp
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_column()` Data-Driven

fill via bare column reference

Use a hidden column to drive fill colors per row.

```
df_bg <- data.frame(
  name = c("Alice", "Bob", "Carol"),
  score = c(95, 72, 88),
  bg = c("green", "red", "yellow"),
  stringsAsFactors = FALSE
)
tt(df_bg, cols = c(name, score)) |>
  tt_column(score, fill = bg)
```

name	score
Alice	95
Bob	72
Carol	88

color via pattern

Use `{col}` pattern to find per-column color columns.

```
df_color <- data.frame(
  a = c(10, 20, 30),
  b = c(40, 50, 60),
  color_a = c("red", "green", "blue"),
  color_b = c("purple", "orange", "black"),
  stringsAsFactors = FALSE
)
tt(df_color, cols = c(a, b)) |>
  tt_column(c(a, b), color = "color_{col}")
```

a	b
10	40
20	50
30	60

bold via bare column reference

Use a boolean column to conditionally bold cells.

```
df_bold <- data.frame(
  item = c("A", "B", "C"),
  value = c(1, 2, 3),
  is_bold = c(TRUE, FALSE, TRUE),
  stringsAsFactors = FALSE
)
tt(df_bold, cols = c(item, value)) |>
  tt_column(value, bold = is_bold)
```

item	value
A	**1**
B	2
C	**3**

italic via pattern

Use `{col}` pattern for per-column italic flags.

```
df_italic <- data.frame(
  x = c("foo", "bar", "baz"),
  italic_x = c(TRUE, FALSE, TRUE),
  stringsAsFactors = FALSE
)
tt(df_italic, cols = x) |>
  tt_column(x, italic = "italic_{col}")
```

x
foo
bar
baz

font_size via pattern

Drive font size from a data column.

```
df_size <- data.frame(
  label = c("Small", "Medium", "Large"),
  size_label = c("8pt", "10pt", "14pt"),
  stringsAsFactors = FALSE
)
tt(df_size, cols = label) |>
  tt_column(label, font_size = "size_{col}")
```

label
Small
Medium
Large

rotate via bare column reference

Drive rotation angle from a data column.

```
df_rot <- data.frame(
  text = c("Up", "Tilt", "Down"),
  angle = c("90deg", "45deg", "-45deg"),
  stringsAsFactors = FALSE
)
tt(df_rot, cols = text) |>
  tt_column(text, rotate = angle)
```

text
Up
Tilt
Down

.missing="ignore" with nonexistent pattern column

Silently skip when a pattern column does not exist.

```r
tt(mt) |>
  tt_column(c(mpg, cyl), color = "color_{col}", .missing = "ignore")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

inset via pattern

Drive cell padding from a data column.

```r
df_inset <- data.frame(
  val = c(1, 2, 3),
  inset_val = c("4pt", "8pt", "16pt"),
  stringsAsFactors = FALSE
)
tt(df_inset, cols = val) |>
  tt_column(val, inset = "inset_{col}")
```

val
1
2
3

`tt_row()`

Header row styling

Style the header row (row 0) with bold, fill, and color.

```r
tt(mt) |>
  tt_row(0, bold = TRUE, fill = "navy", color = "white")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Single data row fill and color

Highlight a single data row.

```
tt(mt) |>
  tt_row(2, fill = "#ffffcc", color = "darkred")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Multiple rows with fill

Style rows 1, 3, and 5 simultaneously.

```
tt(mt) |>
  tt_row(c(1, 3, 5), fill = "#e0e0ff")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

italic and font_size on a row

Apply italic and custom font size to a row.

```r
tt(mt) |>
  tt_row(1, italic = TRUE, font_size = "8pt")
```

mpg	cyl	disp
21	*6*	*160*
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

align override on a row

Override alignment for a specific row.

```r
tt(mt) |>
  tt_row(2, align = "center")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

rotate on a data row

Rotate all content in a data row.

```r
tt(mt) |>
  tt_row(1, rotate = 45)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258

mpg	cyl	disp
18.7	8	360

inset on a row

Apply extra padding to a row.

```
tt(mt) |>
  tt_row(3, inset = "10pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

hline_above and hline_below

Add horizontal lines around a row.

```
tt(mt) |>
  tt_row(3, hline_above = TRUE, hline_below = "2pt + red")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_cell()`

fill and bold on a cell

Highlight a specific cell with fill and bold.

```r
tt(mt) |>
  tt_cell(1, 1, fill = "yellow", bold = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

italic, color, and font_size on a cell

Apply multiple styles to a single cell.

```r
tt(mt) |>
  tt_cell(2, 2, italic = TRUE, color = "red", font_size = "14pt")
```

mpg	cyl	disp
21	6	160
21	*6*	160
22.8	4	108
21.4	6	258
18.7	8	360

align override on a cell

Override alignment for one cell.

```r
tt(mt) |>
  tt_cell(1, 1, align = "right")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258

mpg	cyl	disp
18.7	8	360

rotate on a cell

Rotate a single cell.

```
tt(mt) |>
  tt_cell(3, 1, rotate = 90)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

inset on a cell

Add extra padding to a specific cell.

```
tt(mt) |>
  tt_cell(2, 3, inset = "12pt")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

colspan=2 with content override

Span a cell across two columns with replacement content.

```
tt(mt) |>
  tt_cell(1, 1, colspan = 2, content = "Combined")
```

mpg	cyl	disp
Combined		6
160	21	6
160	22.8	4
108	21.4	6
258	18.7	8
360		

rowspan=3 with content override

Span a cell across three rows with replacement content.

```
tt(mt) |>
  tt_cell(1, 1, rowspan = 3, content = "Grouped")
```

mpg	cyl	disp
Grouped	6	160
	21	6
	160	22.8
4	108	21.4
6	258	18.7
8	360	

content override with styling

Replace cell content and apply formatting.

```
tt(mt) |>
  tt_cell(1, 1, content = "CUSTOM", bold = TRUE, color = "blue", fill = "#f0f0f0")
```

mpg	cyl	disp
CUSTOM	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Header cell styling

Style a cell in the header row (row 0).

```
tt(mt) |>
  tt_cell(0, 1, bold = TRUE, color = "red", fill = "lightyellow")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_hline()` and `tt_vline()`

Custom hline at position 3

Add a horizontal line below the third data row.

```
tt(mt) |>
  tt_hline(3)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Hline with styled stroke

Add a blue 2pt horizontal line.

```
tt(mt) |>
  tt_hline(2, stroke = "2pt + blue")
```

mpg	cyl	disp
21	6	160

mpg	cyl	disp
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Partial hline with start/end

A horizontal line spanning only some columns.

```
tt(mt) |>
  tt_hline(2, start = 0, end = 2)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Vline after column 1

Add a vertical line after the first column.

```
tt(mt) |>
  tt_vline(1)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Vline with styled stroke

Add a gray vertical line.

```
tt(mt) |>
  tt_vline(1, stroke = "1pt + gray")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Partial vline with start/end

A vertical line spanning only some rows.

```
tt(mt) |>
  tt_vline(1, start = 1, end = 5)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`tt_pack_rows()`

Single group with default indent and bold

Pack rows into one labeled group.

```
tt(mt10) |>
  tt_pack_rows("Group A", 1, 5)
```

mpg	cyl	disp
Group A		
21	6	160
21	6	160

mpg	cyl	disp
22.8	4	108
21.4	6	258
18.7	8	360
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

Multiple chained groups

Chain multiple pack_rows calls.

```
tt(mt10) |>
  tt_pack_rows("Group A", 1, 5) |>
  tt_pack_rows("Group B", 6, 10)
```

mpg	cyl	disp
Group A		
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360
Group B		
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

index parameter

Define multiple groups with a named vector.

```r
tt(mtl0) |>
  tt_pack_rows(index = c("First Five" = 5, "Last Five" = 5))
```

mpg	cyl	disp
First Five		
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360
Last Five		
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

indent=FALSE

Disable indentation of grouped rows.

```r
tt(mtl0) |>
  tt_pack_rows("No Indent Group", 1, 5, indent = FALSE)
```

mpg	cyl	disp
No Indent Group		
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360
18.1	6	225
14.3	8	360
24.4	4	146.7

mpg	cyl	disp
22.8	4	140.8
19.2	6	167.6

bold=FALSE

Use a non-bold group label.

```
tt(mt10) |>
  tt_pack_rows("Light Label", 1, 5, bold = FALSE)
```

mpg	cyl	disp
Light Label		
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

Styled group labels

Group labels with fill, italic, and hline_below.

```
tt(mt10) |>
  tt_pack_rows("4 Cylinders", 1, 5, fill = "#e6f3ff", italic = TRUE, hline_below
= TRUE) |>
  tt_pack_rows("6 Cylinders", 6, 10, fill = "#e6f3ff", italic = TRUE, hline_below
= TRUE)
```

mpg	cyl	disp
4 Cylinders		
21	6	160

mpg	cyl	disp
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360
6 Cylinders		
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

Group label with color and alignment

```
tt(mt10) |>
  tt_pack_rows("Centered Group", 1, 5, align = "center", color = "blue", bold
= FALSE)
```

mpg	cyl	disp
Centered Group		
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

Basic `tt_header_above()`

Single spanning header

Add one spanning header group above the columns.

```
tt(mt6) |>
  tt_header_above(c("All Columns" = 6))
```

All Columns					
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Multiple groups with default line_sep

Group columns into multiple labeled spans.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2))
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Check that references still work.

```r
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_cell(row = 3, column = 3, color = 'red')
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Groups with empty spacer

Use a space label for ungrouped columns.

```r
tt(mt6) |>
  tt_header_above(c(" " = 1, "Performance" = 2, "Characteristics" = 3))
```

	Performance		Characteristics		
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

line_sep=NULL (no separator)

Disable separator between header groups.

```r
tt(mt6) |>
   tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2), line_sep
= NULL)
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

line_sep="20pt" (wider separator)

Use wider separator between header groups.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2), line_sep
= "20pt")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Custom styling (bold=FALSE, color, fill, align)

Override the default bold and apply colors.

```
tt(mt6) |>
  tt_header_above(
    c("Performance" = 2, "Engine" = 2, "Design" = 2),
    bold = FALSE,
    color = "white",
    fill = "navy",
    align = "left"
  )
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

line=FALSE

Disable the horizontal line below the spanning header.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2), line = FALSE)
```

Performance		**Engine**		**Design**	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

italic + font_size

Use the `italic` and `font_size` parameters.

```
tt(mt6) |>
  tt_header_above(
    c("Performance" = 2, "Engine" = 2, "Design" = 2),
    italic = TRUE, font_size = "14pt"
  )
```

Performance		***Engine***		***Design***	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

rotate

Rotate the header_above text.

```
tt(mt4) |>
  tt_header_above(c("AB" = 2, "CD" = 2), rotate = 45, line_sep = NULL)
```

AB		CD	
mpg	cyl	disp	hp
21	6	160	110
21	6	160	110
22.8	4	108	93
21.4	6	258	110
18.7	8	360	175

inset

Custom padding on the header_above row.

```
tt(mt6) |>
  tt_header_above(
    c("Performance" = 2, "Engine" = 2, "Design" = 2),
    inset = "12pt", fill = "#f0f0f0"
  )
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Stacked Headers

Two stacked header rows (no separator)

Add two levels of spanning headers without separators.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2), line_sep =
NULL) |>
  tt_header_above(c("Car Metrics" = 6))
```

Car Metrics					
Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Three stacked headers with different styling (no separator)

Multiple header levels with distinct visual styles.

```
tt(mt6, align = 'center') |>
  tt_header_above(c("A" = 2, "B" = 2, "C" = 2)) |>
```

```
  tt_header_above(c("Left" = 2, "Right" = 4), fill = "lightgrey") |>
  tt_header_above(c("All" = 6), fill = "navy", color = "white")
```

All					
Left		Right			
A		B		C	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Two stacked headers with separator on innermost

Stacked headers where the innermost header has separators.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_header_above(c("Car Metrics" = 6))
```

Car Metrics					
Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Three stacked headers with separators

Three stacked levels where the innermost has separators.

```r
tt(mt6) |>
  tt_header_above(c("A" = 2, "B" = 2, "C" = 2)) |>
  tt_header_above(c("Left" = 3, "Right" = 3), fill = "lightgray") |>
  tt_header_above(c("All" = 6), fill = "navy", color = "white")
```

All					
Left			Right		
A		B		C	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Targeting Headers with `tt_row()` (Negative Indices)

Use `tt_row()` with negative row indices to style `tt_header_above()` rows. Row −1 is the innermost (closest to the main header), −2 is the next up, etc.

Italic via tt_row(-1)

Override the innermost header_above row with italic.

```r
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_row(-1, italic = TRUE)
```

Performance		***Engine***		***Design***	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44

Performance		*Engine*		*Design*	
mpg	qsec	cyl	disp	hp	wt
18.1	20.22	6	225	105	3.46

Fill via tt_row(-1)

Apply a fill color to the innermost header_above row.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_row(-1, fill = "#cce5ff")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Override bold=FALSE via tt_row(-1)

The header_above defaults to bold=TRUE. Override it with `tt_row()`.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_row(-1, bold = FALSE, color = "navy")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
18.1	20.22	6	225	105	3.46

Color + font_size via tt_row(-1)

Combine color and font_size overrides.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_row(-1, color = "red", font_size = "14pt")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Style different stacked rows independently

Target each stacked header_above row by its negative index.

```
tt(mt6) |>
  tt_header_above(c("A" = 2, "B" = 2, "C" = 2)) |>
  tt_header_above(c("All" = 6)) |>
  tt_row(-1, fill = "#ffffcc", italic = TRUE) |>
  tt_row(-2, fill = "navy", color = "white")
```

All					
A		*B*		*C*	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32

All					
A		*B*		*C*	
mpg	qsec	cyl	disp	hp	wt
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Three stacked rows with individual styles

Each header level gets a distinct visual treatment.

```
tt(mt6) |>
  tt_header_above(c("A" = 2, "B" = 2, "C" = 2), line_sep = NULL) |>
  tt_header_above(c("Left" = 3, "Right" = 3), line_sep = NULL) |>
  tt_header_above(c("Title" = 6)) |>
  tt_row(-1, fill = "#e8f5e9") |>
  tt_row(-2, fill = "#e3f2fd") |>
  tt_row(-3, fill = "#fce4ec", italic = TRUE)
```

Title					
Left			**Right**		
A		**B**		**C**	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Targeting Headers with `tt_cell()` (Negative Indices)

Use `tt_cell()` with negative row indices to style individual header groups. The column is automatically normalized to the start of the group it falls within.

Fill on one group

Apply fill to just the “Engine” group (cols 3-4).

```r
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_cell(-1, 3, fill = "lightyellow")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Color on one group

Apply red text to the “Design” group (cols 5-6).

```r
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_cell(-1, 5, color = "red")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Content override on a group

Replace the label of a header_above group.

```r
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_cell(-1, 3, content = "Motor")
```

Performance		Motor		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Column normalization: any column in group hits the same cell

Targeting column 1 or column 2 both hit the “Performance” group.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_cell(-1, 2, fill = "lightcoral", italic = TRUE)
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Style multiple groups independently

Give each header group a distinct appearance.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_cell(-1, 1, fill = "#cce5ff") |>
  tt_cell(-1, 3, fill = "#d4edda") |>
  tt_cell(-1, 5, fill = "#f8d7da")
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

tt_cell on outer stacked header

Target a group in an outer (non-innermost) header_above row.

```
tt(mt6) |>
  tt_header_above(c("A" = 2, "B" = 2, "C" = 2), line_sep = NULL) |>
  tt_header_above(c("Left" = 3, "Right" = 3), line_sep = NULL) |>
  tt_cell(-2, 4, fill = "lightyellow", italic = TRUE)
```

Left			*Right*		
A		**B**		**C**	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Content + styling override on stacked header

Replace label and style a group in a stacked configuration.

```
tt(mt6) |>
  tt_header_above(c("A" = 2, "B" = 2, "C" = 2), line_sep = NULL) |>
  tt_header_above(c("Summary" = 6)) |>
  tt_cell(-2, 1, content = "Overview", color = "navy", bold = FALSE) |>
  tt_cell(-1, 3, fill = "#fff3cd")
```

Overview					
A		**B**		**C**	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Combined: tt_row + tt_cell on Header Above

Row-level style + cell-level override

`tt_row(-1)` sets a base style; `tt_cell(-1, ...)` overrides for one group.

```
tt(mt6) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Design" = 2)) |>
  tt_row(-1, fill = "#e0e0e0", italic = TRUE) |>
  tt_cell(-1, 3, fill = "lightyellow", italic = FALSE)
```

Performance		**Engine**		***Design***	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

header_spec defaults + tt_row + tt_cell precedence

The header_spec provides base defaults (bold=TRUE, align=center). `tt_row` overrides at row level, `tt_cell` overrides at cell level. Later calls win.

```
tt(mt6) |>
  tt_header_above(
```

```
    c("Performance" = 2, "Engine" = 2, "Design" = 2),
    color = "gray"
  ) |>
  tt_row(-1, color = "blue") |>
  tt_cell(-1, 5, color = "red", bold = FALSE)
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Full example: stacked headers with mixed targeting

Comprehensive example combining header_above params, tt_row, and tt_cell.

```
tt(mt6) |>
  tt_header_above(
    c("Performance" = 2, "Engine" = 2, "Design" = 2),
    italic = TRUE
  ) |>
  tt_header_above(
    c("Car Data" = 6),
    fill = "navy", color = "white"
  ) |>
  tt_row(-1, fill = "#f0f0f0") |>
  tt_cell(-1, 1, fill = "#cce5ff") |>
  tt_cell(-1, 3, fill = "#d4edda") |>
  tt_cell(-1, 5, fill = "#f8d7da") |>
  tt_cell(-2, 1, content = "All Car Metrics")
```

All Car Metrics					
Performance		***Engine***		***Design***	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62

All Car Metrics					
Performance		***Engine***		***Design***	
mpg	qsec	cyl	disp	hp	wt
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

Precedence / Last-Write-Wins

Row after column (row wins)

When row styling is applied after column styling, the row's fill wins on conflicting attributes.

```
tt(mt) |>
  tt_column(mpg, fill = "lightblue") |>
  tt_row(2, fill = "lightyellow")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Column after row (column wins)

When column styling is applied after row styling, the column's fill wins on conflicting attributes.

```
tt(mt) |>
  tt_row(2, fill = "lightyellow") |>
  tt_column(mpg, fill = "lightblue")
```

mpg	cyl	disp
21	6	160

mpg	cyl	disp
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Cell overrides both row and column

Cell-level styling always takes highest precedence when applied last.

```
tt(mt) |>
  tt_column(mpg, fill = "lightblue") |>
  tt_row(2, fill = "lightyellow") |>
  tt_cell(2, 1, fill = "lightgreen")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Non-conflicting attributes merge

Different attributes from column, row, and cell merge together.

```
tt(mt) |>
  tt_column(mpg, bold = TRUE) |>
  tt_row(2, fill = "lightyellow") |>
  tt_cell(2, 1, italic = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Data-driven column after row

Data-driven column formatting applied after row styling.

```
df_prec <- data.frame(
  val = c(10, 20, 30, 40, 50),
     bg_val   =   c("lightcoral",   "lightgreen",   "lightblue",   "lightyellow",
"lightpink"),
  stringsAsFactors = FALSE
)
tt(df_prec, cols = val) |>
  tt_row(2, fill = "white") |>
  tt_column(val, fill = bg_val)
```

val
10
20
30
40
50

NA Handling

NA values in data cells

Table with NA values in the data.

```
df_na <- data.frame(
  a = c(1, NA, 3),
  b = c(NA, "hello", NA),
  c = c(TRUE, FALSE, NA),
  stringsAsFactors = FALSE
)
tt(df_na)
```

a	b	c
1	-	TRUE
-	hello	FALSE
3	-	-

NA in data-driven style columns

NA values in columns used for data-driven styling.

```
df_na_style <- data.frame(
  val = c(10, 20, 30),
  bg = c("lightblue", NA, "lightyellow"),
  stringsAsFactors = FALSE
)
tt(df_na_style, cols = val) |>
  tt_column(val, fill = bg)
```

val
10
20
30

Complex Compositions

Combined alignment with rotation

Use combined alignment like `"left + bottom"` with rotation.

```
tt(mt) |>
  tt_row(0, rotate = -45, align = "left + bottom")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Full-featured table

Combine header_above, pack_rows, hlines, vlines, column, row, and cell styles.

```
tt(mtcars[1:10, 1:6]) |>
  tt_header_above(c("Performance" = 2, "Engine" = 2, "Other" = 2)) |>
  tt_pack_rows("Group 1", 1, 5) |>
```

```
  tt_pack_rows("Group 2", 6, 10) |>
  tt_hline(1) |>
  tt_vline(2, stroke = "1pt + gray") |>
  tt_column(mpg, bold = TRUE, color = "navy") |>
  tt_row(0, fill = "#e0e0e0") |>
  tt_cell(1, 1, fill = "lightyellow") |>
  tt_style(stroke = TRUE, inset = "4pt")
```

Performance		Engine		Other	
mpg	cyl	disp	hp	drat	wt
Group 1					
21	6	160	110	3.9	2.62
21	6	160	110	3.9	2.875
22.8	4	108	93	3.85	2.32
21.4	6	258	110	3.08	3.215
18.7	8	360	175	3.15	3.44
Group 2					
18.1	6	225	105	2.76	3.46
14.3	8	360	245	3.21	3.57
24.4	4	146.7	62	3.69	3.19
22.8	4	140.8	95	3.92	3.15
19.2	6	167.6	123	3.92	3.44

pack_rows and header_above combined

Use both row grouping and spanning headers.

```
tt(mt10) |>
  tt_header_above(c("A" = 1, "B" = 2)) |>
  tt_pack_rows("First", 1, 5) |>
  tt_pack_rows("Second", 6, 10) |>
  tt_style(stroke = TRUE)
```

A	B	
mpg	cyl	disp
First		
21	6	160
21	6	160
22.8	4	108

A	B	
mpg	cyl	disp
21.4	6	258
18.7	8	360
Second		
18.1	6	225
14.3	8	360
24.4	4	146.7
22.8	4	140.8
19.2	6	167.6

Multiple tt_column calls on same column (last wins)

The last call's conflicting attributes take precedence.

```
tt(mt) |>
  tt_column(mpg, color = "red", bold = TRUE) |>
  tt_column(mpg, color = "blue")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

colspan and rowspan in same table

Use both column spanning and row spanning cells.

```
df_span <- data.frame(
  a = c("r1", "r2", "r3", "r4"),
  b = c("x", "y", "z", "w"),
  c = c(1, 2, 3, 4),
  d = c(5, 6, 7, 8),
  stringsAsFactors = FALSE
)
tt(df_span) |>
```

```
  tt_cell(1, 1, rowspan = 2, content = "Rows 1-2") |>
  tt_cell(3, 2, colspan = 2, content = "Cols 2-3")
```

a	b	c	d
Rows 1-2	x	1	5
	r2	y	2
6	r3	Cols 2-3	
3	7	r4	w
4	8		

Color formats

Test different color specification formats.

```
df_colors <- data.frame(
  hex = c("Hex color", "Another hex", "Third hex"),
  named = c("Named Typst", "Also named", "Yet another"),
  rcolor = c("R color", "Another R", "Third R"),
  stringsAsFactors = FALSE
)
tt(df_colors) |>
  tt_column(hex, color = "#ff5733") |>
  tt_column(named, color = "eastern") |>
  tt_column(rcolor, color = "darkgreen") |>
  tt_cell(1, 1, fill = "#e0f0ff") |>
  tt_cell(2, 2, fill = "aqua") |>
  tt_cell(3, 3, fill = "lavender")
```

hex	named	rcolor
Hex color	Named Typst	R color
Another hex	Also named	Another R
Third hex	Yet another	Third R

Custom pattern

```
tt(mt6) |>
  tt_header_above(c("Performance"=2,"Engine"=2,"Design"=2)) |>
  tt_cell(row=1,column=3, fill='pat')
```

Performance		Engine		Design	
mpg	qsec	cyl	disp	hp	wt
21	16.46	6	160	110	2.62
21	17.02	6	160	110	2.875
22.8	18.61	4	108	93	2.32
21.4	19.44	6	258	110	3.215
18.7	17.02	8	360	175	3.44
18.1	20.22	6	225	105	3.46

`booktabs` Argument

Default booktabs=TRUE

Default table with three horizontal rules and no cell borders.

```
tt(mt)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

booktabs=FALSE (grid style)

Grid-style table with borders on all cells.

```
tt(mt, booktabs = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

booktabs=FALSE with header styling

Grid table combined with header row styling.

```
tt(mt, booktabs = FALSE) |>
  tt_row(0, bold = TRUE, fill = "#e0e0e0")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

booktabs=FALSE with header_above

Grid table with a spanning header.

```
tt(mt, booktabs = FALSE) |>
  tt_header_above(c("Performance" = 2, "Engine" = 1))
```

Performance		Engine
mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

`stroke = FALSE` (Line Suppression)

Suppress top booktabs rule

Remove the top horizontal rule using `tt_hline(0, stroke = FALSE)`.

```
tt(mt) |>
  tt_hline(0, stroke = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Suppress mid booktabs rule

Remove the mid rule (below header) using `tt_row(1, hline_above = FALSE)`.

```
tt(mt) |>
  tt_row(1, hline_above = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Suppress bottom booktabs rule

Remove the bottom rule using `tt_hline()` at the last position.

```
tt(mt) |>
  tt_hline(nrow(mt) + 1, stroke = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Suppress all three booktabs rules

Remove all default rules for a minimal table.

```r
tt(mt) |>
  tt_hline(0, stroke = FALSE) |>
  tt_hline(1, stroke = FALSE) |>
  tt_hline(nrow(mt) + 1, stroke = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Suppress mid rule and add custom replacement

Replace the default mid rule with a colored line.

```r
tt(mt) |>
  tt_row(1, hline_above = FALSE) |>
  tt_hline(1, stroke = "2pt + blue")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

stroke=FALSE on individual cells

Explicitly remove borders from specific cells in a grid table.

```r
tt(mt, booktabs = FALSE) |>
  tt_cell(2, 2, stroke = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160

mpg	cyl	disp
22.8	4	108
21.4	6	258
18.7	8	360

stroke=FALSE on a row in a grid table

Remove borders from an entire row in a grid table.

```
tt(mt, booktabs = FALSE) |>
  tt_row(3, stroke = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

hline_above=FALSE on a grid table row

Suppress the border above a specific row in a grid table.

```
tt(mt, booktabs = FALSE) |>
  tt_row(3, hline_above = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

tt_style(stroke=TRUE) overrides booktabs

Setting `tt_style(stroke = TRUE)` on a booktabs table switches to grid and suppresses the default rules.

```r
tt(mt) |>
  tt_style(stroke = TRUE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

Override a booktabs rule with custom stroke

Replace the top rule with a thick red line.

```r
tt(mt) |>
  tt_hline(0, stroke = "3pt + red")
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258
18.7	8	360

booktabs=FALSE with vline(stroke=FALSE)

Suppress a vertical line in a grid table.

```r
tt(mt, booktabs = FALSE) |>
  tt_vline(1, stroke = FALSE)
```

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108
21.4	6	258

mpg	cyl	disp
18.7	8	360

styling of tt_pack_rows

```
dat <- data.frame(
  group = c("Group A", "Group A", "Group B", "Group B"),
  val = 1:4
)
dat |>
  tt() |>
  tt_pack_rows(index = table(dat$group),
               hline_below = '0.25pt+gray', italic = TRUE, color = 'orange') |>
  tt_row(0:4) |>
  tt_widths(1,1)
```

group	val
Group A	
Group A	1
Group A	2
Group B	
Group B	3
Group B	4